

GATE QUESTION Q36

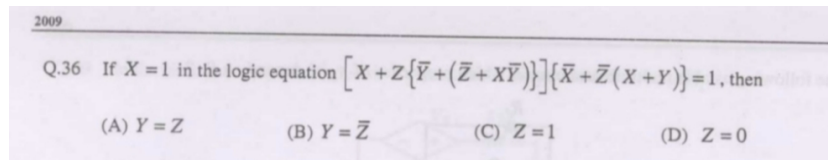


Figure 1: GATE Question Q36

Contents

1 Components

Component	Specification	Quantity
Standard Breadboard	1 Arduino Uno	1
1 IC 7447	BCD to 7 Segment Decoder	1
Common Anode	1 Jumper Wires	1
As Required		

1.1 Breadboard

The breadboard is used to make temporary circuit connections.

1.2 Arduino Uno

Arduino Uno is used to implement the K-Map expression.

1.3 IC 7447

The 7447 is a BCD to Seven Segment Decoder Driver.

1.4 Seven Segment Display

The display is used to show the output of the Boolean expression.

2 Boolean Simplification

The simplified K-Map expression is

[A=]

The remaining outputs are

[B=0, C=0, D=0]

When W=0 and Z=0,

[A=1]

Hence the display shows 1.

3 Hardware Connections

Arduino Pin	Function heightD6
W D7	X D8
Y D9	Z D2
A (7447) D3	B (7447) D4
C (7447) D5	D (7447) D13
Clock height	

4 Software Implementation

```
#include <Arduino.h>

int W,X,Y,Z;
int D,C,B,A;

void disp_kmap(int D,int C,int B,int A)
{
digitalWrite(2,A);
digitalWrite(3,B);
digitalWrite(4,C);
digitalWrite(5,D);
}

void setup()
{
pinMode(2,OUTPUT);
pinMode(3,OUTPUT);
pinMode(4,OUTPUT);
pinMode(5,OUTPUT);

pinMode(6,INPUT);
pinMode(7,INPUT);
pinMode(8,INPUT);
pinMode(9,INPUT);

pinMode(13,OUTPUT);
}

void loop()
{
```

```
digitalWrite(13,HIGH);  
delay(10);  
digitalWrite(13,LOW);  
  
W = digitalRead(6);  
X = digitalRead(7);  
Y = digitalRead(8);  
Z = digitalRead(9);  
  
A = !W & !Z;  
B = 0;  
C = 0;  
D = 0;  
  
disp_kmap(D,C,B,A);  
delay(1000);  
}
```

5 Output

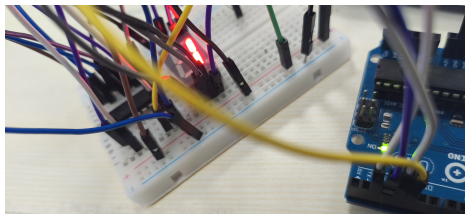


Figure 2: Output Showing 1

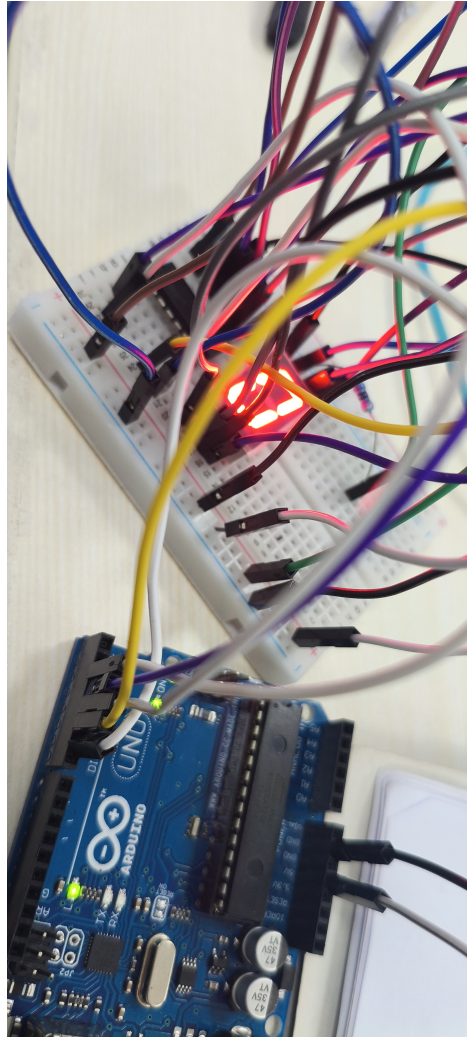


Figure 3: Output Showing 0

6 Result

The Karnaugh Map expression was implemented using Arduino Uno and IC 7447.

[A=]

For W=0 and Z=0,

[A=1]

Therefore the seven segment display shows 1.